Supplementary Material

Identifiability and endpoint reconstruction of time-dependent Jacobi Hamiltonians from projective trajectory jets

Alexander M. Tishin

The notation follows the article. All numerical trajectories are synthetic. The code archive provides the stated inputs, stored trial arrays and recomputation scripts. Local information bounds, empirical errors and noiseless ambiguities describe different experiments.

Supplemental Note 1. Full partial jets and hidden layers

Set $t_0 = 0$ and subtract $E_0(t)I$, so $E_0 = 0$. Write coefficients without derivative factorials as $g_j(t) = \sum_{q\geq 0} G_{j,q}\, t^q$ and $E_n(t) = \sum_{q\geq 0} D_{n,q}\, t^q$. The Schrödinger recurrence and series division define every formal projective coefficient $Z_{n,m} = [t^m]z_n$ from finitely many Hamiltonian coefficients. Its leading term is

$$Z_{n,n} = \frac{(-i)^n}{\hbar^n n!} \prod_{j<n} G_{j,0}. \tag{S1}$$

Thus $z_1, \dots, z_K$ fix $G_{0,0}, \dots, G_{K-1,0}$, successively and uniquely within the positive-coupling convention.

At total order $m \geq n+1$, two fresh real coefficients enter $Z_{n,m}$ with nonzero coefficients

$$\begin{aligned} \frac{\partial Z_{n,m}}{\partial G_{n-1,m-n}} &= \frac{(-i)^n \prod_{j<n-1} G_{j,0}}{\hbar^n (n-1)!\, m} =: \alpha_{n,m}, \\ \frac{\partial Z_{n,m}}{\partial D_{n,m-n-1}} &= \frac{(-i)^{n+1} \prod_{j<n} G_{j,0}}{\hbar^{n+1} n!\, m} =: \beta_{n,m}. \end{aligned} \tag{S2}$$

To see the first identity, the fresh coupling acts on the leading coefficient of c_{n-1} and is integrated once. The fresh diagonal acts on the leading coefficient of c_n. Any later return or extra step raises the total order. Neither coefficient affects another observed $Z_{j,m}$: reaching a different site requires an extra step, including returns to the reference in the denominator. The effects are affine at this order, and

$\beta_{n,m}/\alpha_{n,m} = -iG_{n-1,0}/(n\hbar)$. Consequently the two real columns are linearly independent in the complex plane.

Prescribe arbitrary real formal jets for all deeper coefficients g_j, $j \geq K$, and E_n, $n > K$, retaining positive onset couplings. Proceed through orders $m = 2,3,\ldots$. At each m, for $1 \leq n \leq \min(K, m-1)$ choose the fresh pair $\left(G_{n-1,m-n}, D_{n,m-n-1}\right)$ by the two-dimensional real system that makes $Z_{n,m}$ equal to the original observed coefficient. Earlier coefficients stay fixed. The preceding support argument makes these systems independent within each order. At $m = n+1$, the endpoint identities fix $G_{n-1,1}$ and $D_{n,0}$ to their original values. The first K complete instantaneous layers therefore remain unchanged.

Borel's extension theorem (Borel, 1895, p. 44, cited in the article) realizes each real formal series as a C^∞ germ. For prescribed Taylor coefficients a_q, the prescribed derivatives are $q!\, a_q$, so the theorem supplies the required real smooth germ. There are finitely many coefficient functions; their positive coupling values ensure positivity in a common neighbourhood by continuity. Finite-dimensional smooth Schrödinger evolution has the prescribed formal recurrence, so every observed jet agrees. This proves Theorem 2. Equality of formal jets does not assert equality of trajectories on an interval. Higher observed derivatives constrain combinations of coefficient derivatives, while the deeper complete instantaneous layers remain free. Analytic or bounded-derivative classes require separate analysis.

A finite-order check changes only instantaneous hidden values in a five-level quadratic baseline, $K = 1,2,3$, and two changes of each first hidden coupling and energy. Matching coefficients through order 14 leaves maximum visible-coefficient residual below 4.2×10^{-17} and the reconstructed first K coupling values, rates and energies unchanged within 4.5×10^{-16}. The compensating Taylor coefficients stay of order one in these tests, with maximum magnitude 3.84. The mathematical conclusion follows from the recursion and Borel extension, rather than this finite-order check.

Supplemental Note 2. Observation-gap witness and derivative bound

Corollary 2 (a derivative-bounded error floor). Fix a constant real Jacobi matrix H_* with positive couplings, projectively centered norm $B = \min_{s\in\mathbb{R}} \| H_* - sI \| > 0$ and target $\theta_* = E_n - E_0 \neq 0$. Let $\mathcal{H}_L$ be all smooth real Jacobi drives with positive couplings and $\| \dot{H}(t) \| \leq L$ on $[0,T]$, $L > 0$, and let

$\theta(H) = E_n(0) - E_0(0)$. For the state-only experiment on $[\tau, T]$, define $R_q = \inf_{\hat{\theta}} \sup_{H \in \mathcal{H}_L} \mathbb{E}_H \left| \hat{\theta} - \theta(H) \right|^q$. Then $R_1 \geq \varepsilon_{\mathrm{w}} |\theta_*|/2$ and $R_2 \geq \varepsilon_{\mathrm{w}}^2 \theta_*^2/4$, where

$$\varepsilon_{\mathrm{w}} = \min\{1/2, L\tau/(K_2 B)\}. \tag{S3}$$

These are lower bounds from a constructed subclass; no general upper bound on ambiguity follows from L alone.

Proof. Replace H_* by its spectral-midpoint scalar shift, which preserves the target and projective trajectory and has norm B. For $0 \leq s < 1$ set $\eta(s) = s\exp[-s/(1-s)]$, extended by zero for $s \geq 1$, and $f(t) = t + \varepsilon_{\mathrm{w}} \tau \eta(t/\tau)$. Writing $u = (1-s)^{-1} \geq 1$ gives $\eta' = e^{1-u}(1 + u - u^2)$ and $\eta'' = e^{1-u}(u^4 - 3u^3)$. The extrema of η' occur at $s = 0$, $u = 3$, or the vanishing endpoint, giving $\sup|\eta'| = 1$ and $\min \eta' = -5e^{-2}$. The constant is $K_2 = \max_{u \in \{(7-\sqrt{13})/2, (7+\sqrt{13})/2\}} e^{1-u} |u^4 - 3u^3|$. The stationary values of η'' occur at $u = \left(7 \pm \sqrt{13}\right)/2$, since its derivative has factor $-u^2(u^2 - 7u + 9)$. Thus $\sup|\eta''| = K_2 \simeq 4.6461$. These exact extremum formulas give $f' \geq 1 - \varepsilon_{\mathrm{w}} \geq 1/2$ and $\| \ddot{\tilde{H}} \| \leq K_2 \varepsilon_{\mathrm{w}} B/\tau \leq L$. The target separation is $\varepsilon_{\mathrm{w}} \theta_*$. The common-data two-point inequality gives the stated expected absolute and squared error floors. Any smaller class containing both witnesses has the same lower bounds.

Without endpoint-size bounds, Proposition 2 also gives unbounded worst-case squared risk. With a derivative bound, Corollary 2 shows that the lower edge of an acquisition window remains relevant even at zero observation noise. The quoted scale is a target error, rather than a standard deviation or a Fisher estimate. For the six-level control, spectral centering gives $B \simeq 2.334$. On $[0,0.24]$, the sufficient envelope $L = \| \dot{H}(0) \| + 0.24 \| \ddot{H} \| \simeq 0.3065$ and $\tau = 0.024$ give $R_1 \geq 2.37 \times 10^{-4}$ for $|\theta_*| = 0.70$. This subclass floor is below the reported finite-window errors; it does not explain their magnitude.

Supplemental Note 3. Derivative frame and established constructions

Canonical endpoint frame

The coordinate result is covariant under fixed unitary changes of ambient coordinates. Assume throughout this section that the Hamiltonian is real Jacobi, with positive accessible couplings in a single time-independent orthonormal ladder basis near endpoint preparation. Under this prior, endpoint

osculating vectors recover accessible ladder vectors with the phases specified in Supplemental Proposition S1, including when their ambient representation is not supplied. The derivative-QR connection of a general smooth curve need not equal the physical Hamiltonian. Define the osculating flag

$$\mathcal{V}_m(t_0) = \mathrm{span}\{\psi, \dot{\psi}, \dots, \psi^{(m)}\}_{t=t_0}. \tag{S4}$$

For the endpoint Jacobi chain, the unique minimal path implies

$$\mathcal{V}_m(t_0) = \mathrm{span}\{e_0, \dots, e_m\}. \tag{S5}$$

Apply complex QR–Gram–Schmidt to the derivative columns $J_M = \left(\psi, \dot{\psi}, \dots, \psi^{(M)}\right)$, with every nonzero QR diagonal chosen positive and real. This convention defines a canonical local frame $Q = (q_0, q_1, \dots)$ and fixes the independent phases of the frame vectors. Let

$$\Gamma = Q^{\dagger}\dot{Q}. \tag{S6}$$

Supplemental Proposition S1 Endpoint osculating connection. At t_0, the canonical frame satisfies $q_n = (-i)^n e_n$ for $0 \le n \le M$, up to the common phase of the Schrödinger lift. For $0 \le n < M$ its frame evolution obeys

$$\dot{q}_n = -\frac{g_{n-1}}{\hbar} q_{n-1} - i\frac{E_n}{\hbar} q_n + \frac{g_n}{\hbar} q_{n+1}, g_{-1} = 0. \tag{S7}$$

Hence

$$\begin{gathered} E_n - E_0 = i\hbar(\Gamma_{nn} - \Gamma_{00})(0 \le n \le M), \\ g_n = \hbar\Gamma_{n+1,n} > 0(0 \le n < M). \end{gathered} \tag{S8}$$

The instantaneous connection in (S8) does not by itself encode the logarithmic coupling rates. For the same QR decomposition, write $r_n(t) = \mathrm{R}_{nn}(t) > 0$. Coupling rates follow from

$$\frac{\dot{g}_{n-1}(t_0)}{g_{n-1}(t_0)} = \frac{1}{n}\frac{d}{dt}\log r_n(t)\Big|_{t=t_0}, 1 \le n \le M. \tag{S9}$$

Thus (S8) and (S9), together, represent all data in Theorem 1. The derivative in (S9) belongs to the QR factor along the trajectory. Gauge properties of the connection and of the QR factor are given below under Gauge and structural assumptions.

The phase-sensitive connection contains information that scalar metric invariants can lose. The phase-resolved connection identifies signed energy offsets under the fixed real-chain assumptions.

For a general moving orthonormal frame, state coordinates obey $i\hbar\dot{b} = (Q^\dagger HQ - i\hbar\Gamma)b$. The connection is not $Q^\dagger HQ$. In particular, with $q_0 = \psi$, the transformed state has only its first component nonzero, and its equation constrains only the first column of that difference. The identification in (S8) is a special endpoint consequence of the fixed real-chain prior. For example, $H_\delta(t) = H(t) + \delta(t)[I - |\psi(t)\rangle\langle\psi(t)|]$ produces the same entire Schrödinger trajectory and QR frame; for $N > 1$ this changes the action on the orthogonal subspace and is not a scalar gauge. Such a generator generally leaves the assumed real fixed-chain class, so it does not contradict Theorem 1.

The operational inputs differ for known and unknown ladder vectors. In an addressable known physical ladder, the reference population and the M complex coherences $c_n c_0^*$, with their time derivatives through $M + 1$, supply the coordinate jet for M complete layers. The acquisition protocol below assumes this access. If the ladder's ambient representation is unknown, we assume access to the full pure-state projector jet in known ambient coordinates through order $M + 1$, or a smooth normalized lift of that jet. Choosing such a lift fixes only its common phase; no observable global phase is additionally required. This is stronger tomographic input than the listed coordinate measurements. The shot count for the known ladder does not implement or bound the cost of unknown-basis recovery. Even with that input, the theorem identifies only the accessible endpoint layers under the independently assumed fixed-basis class; it does not determine the operator on the entire unresolved complement or its future time law.

Scalar metric speed supplies less information than the phase-resolved endpoint jet; its value $\Delta H/\hbar$ depends only on the first coupling at preparation.

Endpoint connection proof

Let

$$J_M(t) = \left(\psi, \dot{\psi}, \ldots, \psi^{(M)}\right) = Q(t)\mathrm{R}(t), \tag{S10}$$

where $Q^\dagger Q = I$ and each nonzero diagonal element of the upper-triangular R is chosen positive and real. At the endpoint, $\psi^{(m)}(t_0)$ has no support above e_m, while the coefficient of e_m is $(-i/\hbar)^m g_0 \cdots g_{m-1}$. Therefore

$$\mathcal{V}_m(t_0) = \mathrm{span}\{e_0, \dots, e_m\}, q_m(t_0) = (-i)^m e_m \tag{S11}$$

up to a common lift phase. The corresponding positive QR diagonal is

$$r_m \equiv \mathsf{R}_{mm} = \frac{1}{\hbar^m} \prod_{j=0}^{m-1} g_j\,(t_0) > 0. \tag{S12}$$

Differentiate $J = QR$ and left-multiply by $Q^\dagger$. With

$$X \equiv Q^\dagger \dot{J} \mathsf{R}^{-1} = \Gamma + \dot{\mathsf{R}}\mathsf{R}^{-1}, \Gamma = Q^\dagger \dot{Q}, \tag{S13}$$

$\dot{\mathsf{R}}\mathsf{R}^{-1}$ is upper triangular and Γ is anti-Hermitian. Hence

$$\begin{gathered} \Gamma_{ij} = X_{ij} (i > j), \\ \Gamma_{ii} = i\mathrm{Im}X_{ii}, \\ \Gamma_{ij} = -\Gamma_{ji}^* (i < j). \end{gathered} \tag{S14}$$

The lower part of X follows directly from support and the unique minimal path. In column n, every term inherited from derivative columns of order at most n has support in $span\{e_0, \dots, e_n\}$. Only $\psi^{(n+1)}$ reaches e_{n+1}, and its coefficient there is $(-i/\hbar)^{n+1} g_0 \cdots g_n$. Since $(\mathsf{R}^{-1})_{nn} = r_n^{-1}$,

$$\begin{gathered} X_{n+1,n} = \frac{\langle q_{n+1}, \psi^{(n+1)} \rangle}{r_n} = \frac{g_n}{\hbar}, \\ X_{in} = 0 (i > n+1). \end{gathered} \tag{S15}$$

For the full diagonal, define $d_n = \langle e_n, \psi^{(n+1)} \rangle / \langle e_n, \psi^{(n)} \rangle$ at the endpoint, with $d_0 = -iE_0/\hbar$. Only derivative columns n and $n-1$ contribute to X_{nn}, so $X_{nn} = d_n - d_{n-1}$. The physical-amplitude version of the recursion (29) of the article gives

$$X_{nn} = n \frac{\dot{g}_{n-1}}{g_{n-1}} - \frac{iE_n}{\hbar} (1 \le n \le M), X_{00} = -\frac{iE_0}{\hbar}. \tag{S16}$$

Since R is upper triangular with positive real diagonal, $\left(\dot{\mathsf{R}}\mathsf{R}^{-1}\right)_{nn} = \dot{r}_n / r_n$. The diagonal of Γ is purely imaginary, so (S16) gives both $\dot{r}_n / r_n = n\dot{g}_{n-1}/g_{n-1}$ and $\Gamma_{nn} = -iE_n/\hbar$. Equations (S13)–(S16) therefore yield (S9) and the frame evolution

$$\dot{q}_n = -\frac{g_{n-1}}{\hbar} q_{n-1} - i\frac{E_n}{\hbar} q_n + \frac{g_n}{\hbar} q_{n+1}, \tag{S17}$$

whenever the neighbouring frame vector exists; the diagonal identity remains valid at the reconstruction boundary even if the next coupling vanishes. Hence (S8) follows. For the Schrödinger lift $q_0 = \psi$, $\Gamma_{00} = -iE_0/\hbar$ at the endpoint. A common lift phase adds the same imaginary scalar to every Γ_{nn} and cancels from their differences. Independent frame phases are fixed locally by the positive-real QR diagonal convention.

Gauge and structural assumptions

This note collects the gauge properties of the canonical derivative frame used in the endpoint osculating-connection result. Differentiating the endpoint-only product formula for the positive QR diagonal as if it held at neighbouring times would be incorrect. A scalar lift phase leaves each $r_n(t)$ unchanged, because derivatives of that phase mix derivative columns by an upper-triangular matrix with unit diagonal, apart from the common unit-modulus multiplier.

A common lift phase $\psi \mapsto e^{i\chi(t)}\psi$ adds the same $i\dot{\chi}$ to all diagonal entries of Γ and therefore cancels from their differences. A fixed unitary change of ambient coordinates transforms J_M and Q together and leaves Γ unchanged. Arbitrary independent time-dependent rephasings of q_n would change Γ and are excluded by the positive-diagonal QR convention. Laboratory time is physical input; arbitrary reparameterizations are not allowed.

The positive QR convention fixes a frame constructed from the trajectory. It does not certify that the unknown physical Hamiltonian belongs to the real Jacobi class in one fixed ladder basis throughout a neighbourhood of the endpoint. Identification with physical endpoint coefficients requires that class as an independent prior. The moving-frame relation above uses $b = Q^\dagger\psi$ and $\Gamma = Q^\dagger\dot{Q}$. The connection alone is not the physical Hamiltonian. A derivative-QR connection can be tridiagonal for a general smooth curve because differentiation raises the osculating flag by at most one level and the connection is anti-Hermitian. Under the fixed real Jacobi assumptions, the special endpoint identities of the main article recover the accessible ladder vectors and coefficients.

Static dictionary and time-dependent basis transport

The endpoint construction applies constrained state-derivative reconstruction to the fixed real-Jacobi class. Its rank-one instantaneous covariance matrix exposes the first coupling at first order; higher

derivatives yield the deeper coefficients and coupling rates with the necessary derivative order. The main article also establishes separated-window ambiguity and a derivative-bounded error floor.

Forward Lanczos and time-dependent Krylov constructions use a known generator. Holomorphic Arnoldi geometry concerns fixed deformations, while classical inverse Jacobi theory uses spectral, moment and boundary-response data. Hamiltonian-learning methods use controlled inputs, observable time traces, sparse priors or continuous measurements. The main article discusses these approaches and their relation to the present phase-resolved endpoint data. The fixed-basis condition is stated above; it is required to identify recovered coefficients with the physical Jacobi Hamiltonian.

The derivative-QR construction is a complex generalized Frenet frame. Wherever the derivative columns remain independent, differentiating their upper-triangular expansion gives

$$\Gamma_{n+1,n}(t) = r_{n+1}(t)/r_n(t). \tag{S18}$$

This identity is a frame fact; the physical endpoint identification additionally uses the fixed real-Jacobi prior. In the static case, set $\mu_k = \langle e_0|(H/\hbar)^k|e_0\rangle$ and $\Delta_n = \det\left[\mu_{j+k}\right]_{j,k=0}^{n}$, with $\Delta_{-1} = 1$. The derivative Gram determinants give

$$r_n^2 = \frac{\Delta_n}{\Delta_{n-1}}, \frac{g_n}{\hbar} = \frac{\sqrt{\Delta_{n+1}\Delta_{n-1}}}{\Delta_n}. \tag{S19}$$

These are classical moment/Lanczos identities, not separate inverse results. The main bibliography credits Gluck for generalized Frenet geometry and Gragg–Harrod and Golub–Meurant for Jacobi/moment methods. In Takahashi–del Campo's moving-basis formulation the effective generator is $Q^\dagger H Q - i\hbar Q^\dagger \dot{Q}$. Here $\Gamma = Q^\dagger \dot{Q}$ is the transport connection and r_{n+1}/r_n is its off-diagonal derivative-frame coefficient. For a static cyclic endpoint the state Krylov chain is the given Jacobi chain. Their counterdiabatic Krylov space is an operator space and serves a different forward-control task.

For the derivative-QR frame the first column is the normalized state itself, so $Q^\dagger \psi = (1,0,\dots)^T$ wherever this frame exists. This coefficient vector cannot be identified with the evolving spread amplitudes in a separately chosen Krylov basis. The transport formula relates the effective generators when the relevant bases are specified. The endpoint flag and static moment identities above are the corresponding restricted dictionaries.

Supplemental Note 4. Numerical methods, nuisance models and error summaries

Expanded nuisance models and numerical precision

The six-level control has endpoint diagonals and couplings

$$E(0) = (0.12,0.37,1.91,0.82,3.44,1.27),\ g(0) = (0.73,1.11,0.58,0.94,0.67). \tag{S20}$$

Each coefficient follows the quadratic law $a(t) = a(0) + \dot{a}(0)t + \ddot{a}(0)t^2/2$, with

$$\begin{aligned} \dot{E} &= (0.07, -0.13,0.05,0.19, -0.11,0.03), \\ \ddot{E} &= (0.02, -0.01,0.04, -0.03,0.01,0.05), \\ \dot{g} &= (-0.19,0.08,0.14, -0.05,0.09), \\ \ddot{g} &= (0.06, -0.02,0.03,0.04, -0.01). \end{aligned} \tag{S21}$$

The observations are $\mathrm{Re}z_1, \mathrm{Im}z_1, \ldots, \mathrm{Re}z_3, \mathrm{Im}z_3$ at 61 uniformly spaced times in $[T/10, T]$, with independent Gaussian component noise σ. The target is the constant shift of $E_3(0)$, equivalent to $E_3 - E_0$ with E_0 fixed. The nested parameter sets are:

- S1: endpoint values $E_1, E_2, E_3, g_0, g_1, g_2$ (6 parameters).
- S2: S1 and their first derivatives (12 parameters).
- S3: endpoint values $E_1, \ldots, E_4, g_0, \ldots, g_3$, and their first and second derivatives (24 parameters).

All listed parameters other than $E_3(0)$ are nuisance. E_0, E_5, g_4 and their derivatives, and every unlisted coefficient, are fixed. S3 is a finite quadratic parameter class, not unrestricted smooth nuisance functions.

For reproducibility, write $H(t) = \sum_{q=0}^{2} H_q\, t^q$, where the entries in H_q are the qth time derivatives divided by $q!$. The state series $c(t) = \sum_{k\geq 0} c_k\, t^k$ and each parameter sensitivity $s_p(t) = \partial_p c(t) = \sum_{k\geq 0} s_{p,k}\, t^k$ obey, in the dimensionless units used here,

$$\begin{aligned} c_0 &= e_0, \quad c_{k+1} = -\frac{i}{k+1} \sum_{q=0}^{\min(k,2)} H_q\, c_{k-q}, \\ s_{p,0} &= 0, \quad s_{p,k+1} = -\frac{i}{k+1} \left[\sum_{q=0}^{\min(k,2)} H_q\, s_{p,k-q} + \sum_{q=0}^{\min(k,2)} \left(\partial_p H_q\right) c_{k-q} \right]. \end{aligned} \tag{S22}$$

For parameter p equal to the qth derivative of a diagonal coefficient, $\partial_p H_q = D_j/q!$; for a coupling derivative it is $B_j/q!$. Other derivative matrices vanish. S3 orders parameters by $q = 0,1,2$, with $E_1, E_2, E_3, E_4, g_0, g_1, g_2, g_3$ within each block. The target is the third entry of the $q = 0$ block, $E_3(0)$. S1 and S2 retain their specified subsets in this ordering. At each sample time,

$$\partial_p z_n = \frac{s_{p,n} c_0 - c_n s_{p,0}}{c_0^2}. \tag{S23}$$

The real Jacobian has 366 rows: the 183 real components ordered by time and then $n = 1,2,3$, followed by the corresponding 183 imaginary components. Let v denote its unnormalized target column and N its nuisance columns individually divided by their Euclidean norms. In high precision the projection is evaluated without discarding singular modes:

$$r = v - N(N^T N)^{-1} N^T v, I_{\text{eff}} = \frac{\| r \|^2}{\sigma^2}, I_{\text{fixed}} = \frac{\| v \|^2}{\sigma^2}. \tag{S24}$$

Normalization changes only the nuisance span. Singular spectra are computed from the full column-normalized Jacobian; the quoted double-precision threshold is diagnostic and is not used to truncate the high-precision projection.

Efficient information projects the target sensitivity off all nuisance columns. The standard-deviation inflation relative to fixed-nuisance information is:

Table S1. Local standard-deviation inflation factors for the stated models.

T	S1	S2	S3
0.12	1.00001	10.0	1700
0.24	1.00002	10.1	1320
0.48	1.00014	10.1	870

The column-normalized Jacobians have full ranks 6, 12 and 24. Full rank establishes local differential identifiability in these finite-dimensional models. It does not guarantee global uniqueness or accurate estimation at finite noise. The smallest normalized S3 singular values are approximately 9.01439×10^{-10}, 7.84131×10^{-9} and 7.48547×10^{-8} for the three windows. The corresponding information is extremely sensitive to errors in the sensitivity columns. Operational identifiability does not follow from a numerical rank test.

Double-precision variational integration is insufficient to determine S3 information reliably in the shorter windows. Tightened integrations produced inflation factors about 736.5, 1278.2 and 870.1. These integrations serve as sensitivity-error diagnostics. A double-precision Taylor recurrence independently reproduces the printed table. The reported S3 factors use a Taylor recurrence for state and sensitivities with exact decimal input coefficients. Calculations at 50 decimal digits/order 50 and 70 digits/order 70 agree to the stored significant digits; no weak sensitivity direction is silently discarded. This is a numerical stability comparison, not a rigorous Taylor-truncation bound. Convergence of state coordinates or a stable computed rank alone does not certify the small residual after nuisance projection.

Finite-window bias and interpretation of estimator errors

For polynomial degree $n + 2$, linear propagation of component noise $\sigma = 10^{-8}$ about the noiseless fit gives scales 0.0707, 0.00446 and 0.000287 at $T = 0.12,0.24,0.48$. For degree $n + 3$ the scales are 0.388, 0.0244 and 0.00154. These scales come from the delta method and are not empirical finite-sample standard deviations. The unregularized ratio estimator, with nondegenerate jointly Gaussian fitted numerator and complex denominator, has generally divergent second moments despite finite first absolute moments.

At $T = 0.24$, the fixed-nuisance scale is 5.31×10^{-5}, the S2 regular locally unbiased bound is 5.33553×10^{-4}, and the S3 bound is 0.0699. A small empirical polynomial median does not violate the S3 bound. That estimator has parameter-dependent finite-window bias and is not locally unbiased in S3. At the noiseless fit, its derivative with respect to the target is 1.018 instead of one; the maximum absolute response to a nuisance coefficient is 0.132 instead of zero. This maximum depends on parameter units and is not an invariant condition number.

At $T = 0.24$, the biases of $-\mathrm{Im}R_2$ and $-\mathrm{Im}R_3$ are 0.0178 and 0.0185. The third-energy estimator subtracts them, leaving only 0.000679. Its small bias includes substantial cancellation rather than uniformly small error in each reconstructed derivative ratio. A noiseless window sweep gives signed third-energy bias -0.000697003, -0.000176281, 0.000679, 0.00204868 and 0.0361476 at $T = 0.12,0.21,0.24,0.27,0.48$. The sign change between 0.21 and 0.24 is specific to this chain.

Monte Carlo uncertainty and reproducibility

For each group of 200 trials, 20,000 bootstrap resamples were generated with seed 2026101003. Window groups were resampled independently. Percentile 95% intervals quantify Monte Carlo uncertainty of the sample median under the empirical distribution; they are not Hamiltonian confidence intervals or a universal window-selection rule.

Table S2. Median absolute third-energy errors and bootstrap percentile intervals, $\sigma = 10^{-8}$ and degree $n + 2$.

T	Median	95% interval
0.12	0.0406	[0.0347, 0.0500]
0.24	0.00308	[0.00244, 0.00355]
0.48	0.0362	[0.03614, 0.03626]

For 0.24 minus 0.12, the difference of medians is -0.0375730, with interval $[-0.0467399, -0.0314153]$. For 0.24 minus 0.48, it is -0.0331339, with interval $[-0.0337563, -0.0326461]$. The tested $T = 0.24$ window performs better than these neighbours for this chain, noise and degree; this is not optimization over arbitrary chains or continuous T.

At $T = 0.24$, degree-$n + 2$ medians for $\sigma = 10^{-10}, 10^{-8}, 10^{-6}$ are 0.00067804, 0.00308 and 0.307752. Degree-$n + 3$ medians are 0.00245532, 0.0169 and 1.58011. Its $\sigma = 10^{-8}$ bootstrap interval is $[0.0136, 0.0200]$. Higher degree does not improve this setting.

The computational files specify the parameter lists, normalized singular spectra, rank thresholds, conditioning, solver controls, and precision comparisons. Reproduction uses extended_fisher.py, high_precision_fisher.py, and bootstrap_windows.py, with the supplied requirements. The bootstrap script reads packaged trial records directly. The source digest and resampling controls are recorded with the bootstrap outputs; the accompanying README specifies the portable commands. The Taylor projection results and the ODE diagnostics are recorded separately. Bootstrap analysis resamples the stored trial arrays.

Nuisance-group and response checks

The group decomposition uses the same six-level laws and 61 coordinate observations. Starting with the quadratic coefficients of $E_1, E_2, E_3, g_0, g_1, g_2$, the added groups are the endpoint values of E_4, g_3,

their first derivatives, then their second derivatives. A Taylor state/sensitivity series through order 70 and an SVD projection without mode deletion reproduce the factors $(55.3,56.5,224,1700)$, $(55.6,57.9,228,1320)$ and $(56.2,61.6,238,870)$ at $T = 0.12,0.24,0.48$. Removing E_4 and its derivatives from S3 gives factors about 58, 64, 79.

The response vector b is the derivative of the noiseless polynomial energy map with respect to all 24 S3 coordinates. With the Gaussian coordinate Fisher matrix I, the local quantities $\sqrt{b^T I^{-1} b}$ at $\sigma = 10^{-8}$ are 0.0707, 0.00446 and 0.000287, matching the corresponding delta-method scales at the quoted precision. If a is the data-gradient of the noiseless map and J its state-sensitivity Jacobian, then $b = J^T a$. With $I = J^T J/\sigma^2$, $\sqrt{b^T I^{-1} b} = \sigma \parallel P_{\mathrm{col}J} a \parallel$, whereas the delta scale is $\sigma \parallel a \parallel$. Their ratios are approximately 1.00, 0.999 and 0.999 for the three windows. Thus almost all local data-gradient norm lies in the sensitivity span. These are linearized response-function diagnostics. The actual noisy ratio has divergent second moments, and the computed b is a noiseless-map derivative. A fixed positive denominator clipping threshold produces finite moments. At $\sigma = 10^{-8}$, for degree $n + 2$ and $T \geq 0.12$, the third onset denominator exceeds 10^3 local component standard deviations, so near-zero events are practically remote in that regime. For degree $n + 3$ at $T = 0.12$, the value is only about 356; shorter tested windows can have values as low as order one, where near-zero denominator events can become relevant. Figure 1 at $T \leq 0.06$ includes the small-denominator regime. Medians remain useful summaries while the unregularized ratio has divergent second moments. This supports the operational relevance of the local diagnostic in the well-separated regime; it does not by itself establish a finite-noise efficiency theorem for a clipped estimator. A finite-noise biased-estimator Cramér–Rao theorem would require a finite-variance estimator and the derivative of its expected value.

Units and acquisition cost

The calculations use $t = \Omega t_{\mathrm{lab}}$ and $H = H_{\mathrm{lab}}/(\hbar\Omega)$. Time-law derivatives transform accordingly. For example, $\Omega = 2\pi \times 1$ MHz maps $T = 0.24$ to 38.2 ns, $T/10$ to 3.82 ns and $\tau_r = 0.002$ to 0.318 ns. The frequency Ω is a chosen conversion scale, rather than a fitted apparatus parameter. The displayed coefficients specify the complete synthetic model. Under ideal pair-projector acquisition the total copy counts scale as the inverse square of a demanded coordinate precision. The stated examples 4.27×10^{14} and 4.27×10^{12} would take about 13.5 years and 49.4 days at a continuous rate of 10^6 copies/s, before calibration, reset and control overhead.

For a clipped local-polynomial estimator, a bias bound $C_b T^r$ and a variance bound $C_v/(ST^{2n+2})$ would imply a balanced mean-squared-error scale $S^{-2r/(2r+2n+2)}$. For root-mean-squared error ε, the corresponding conditional count scaling is $S \propto \varepsilon^{-2(r+n+1)/r}$. This algebra is conditional on those uniform bounds and on the measurement model. It supplies neither an achievable universal exponent nor a bound for an unclipped Gaussian ratio.

Fixed-nuisance distinguishability proof

The path argument of Corollary 1, with $M = n$, gives $\tilde{z}_n - z_n = (-i)^{n+1}\delta\kappa_n\tau^{n+1} + O(\tau^{n+2})$. Every other observed coordinate changes at order at least $n + 2$. A changed path must visit the shifted diagonal and then reach a different level. For fixed sampling fractions, summing the squared real and imaginary differences therefore gives $\| \tilde{\mu} - \mu \|_2^2 = \delta^2\kappa_n^2 T^{2n+2} w_n[1 + O(T)]$, proving the first identity in (16) of the article. The remainder constants depend on the fixed Hamiltonians and are not uniform over arbitrary smooth drives.

For Gaussian observations with covariance $\sigma^2 I$, the equal-prior likelihood-ratio test projects onto the difference of the two mean vectors and thresholds at their midpoint. Their normalized separation is $\mathcal{D}$, giving exact error $\Phi(-\mathcal{D}/2)$. Differentiating the leading signal with respect to the constant diagonal shift at zero gives the stated fixed-nuisance Fisher information. Its inverse supplies (17) of the article under the usual regularity and local-unbiasedness conditions. Unknown nuisance parameters reduce information through their Schur complement, or the equivalent sensitivity projection. This argument assumes additive coordinate noise and does not identify it with a particular shot-noise apparatus. Remainder constants are not uniform over arbitrary smooth drives.

Supplemental Note 5. Equal-copy acquisition designs

All designs below estimate the same endpoint targets $g_2(0)$, $E_3(0) - E_0(0)$ and $\dot{g}_2(0)/g_2(0)$ in the same 24-parameter S3 quadratic class. All other coefficients are supplied. The Fisher calculation uses exact pure preparations, calibrated pair quadratures, reference-population discrimination and independent multinomial/binomial counts. A total budget $B_{\mathrm{acq}} = 4.27 \times 10^{12}$ is equally allocated over 61 times and the listed settings. The later windows require longer evolution per copy. No calibration copies, reset durations or pulse overhead are included in this ideal information comparison. Bounds are local and apply to regular locally unbiased estimators. They are not achieved estimator errors.

Table S3. Ideal equal-copy local standard-deviation bounds for the same endpoint targets: $\Delta E_3 = E_3(0) - E_0(0)$ and $\chi_2 = \dot{g}_2(0)/g_2(0)$; s is the settings count. All windows have width 0.216. Later-time designs use the same endpoint preparation, not a new state preparation at the window center.

Design	Window	s	g_2	ΔE_3	χ_2
Early, z_1–z_3	[0.024,0.240]	7	0.137	24.2	7.30
Early, z_1–z_4	[0.024,0.240]	9	0.0349	1.68	0.852
Later, z_1–z_4	[0.192,0.408]	9	0.00716	0.173	0.103
Later, z_1–z_4	[0.392,0.608]	9	0.00360	0.0476	0.0337
Two sites 0,2	[0.024,0.240]	8	3.32×10^{-5}	0.0112	0.00143

The two-site design observes the pair $0 \leftrightarrow 1$ from site 0 and pairs $2 \leftrightarrow 1,3$ from site 2. The four-level single-preparation design observes pairs $0 \leftrightarrow 1,2,3,4$. Adding z_4 removes a substantial weak-nuisance contribution, and later windows further improve information. Under exact preparation, among these specified designs the two-site input is most informative for each endpoint target, but arbitrary coherent probes, optimized windows, unequal allocations and other estimators are not optimized here.

With independent equal Gaussian component noise, the efficient energy scale per unit σ for the endpoint target is 6.99×10^6 for early z_1–z_3, 2.95×10^5 for early z_1–z_4, 2.99×10^4 for the window centered at 0.3, and 7.93×10^3 for the window centered at 0.5. Estimating the different target $E_3(0.3) - E_0(0.3)$ or $E_3(0.5) - E_0(0.5)$ gives scales 1.75×10^3 and 294, respectively. Those reductions must not be represented as same-target endpoint gains. Equal Gaussian coordinate noise also does not charge the extra quadrature settings; the multinomial table does.

The weak early-three-level multinomial design was independently checked at 40-digit/order-50 and 60-digit/order-70 precision; its energy scale $\mathrm{sd}\sqrt{B_{\mathrm{acq}}}$ agrees at the reported precision and is 5.00319×10^7. Double-precision ODE sensitivities yield a value differing by about 0.6% despite small component errors, whereas the other four design comparisons differ by less than 10^{-8} in the energy scale. This exposes the sensitivity of the weak direction and motivates the high-precision calculation.

The state and sensitivity Taylor recurrence through order 70 was independently compared with DOP853 integration over times through 0.716. Maximum component differences were below $3 \times$

10^{-14} for both preparations. Singular modes are retained; column normalization changes the conditioning of the calculation, not the nuisance span. The archive gives unrounded bounds and singular values. Extremely weak directions remain sensitive to floating-point perturbations, so the table uses rounded scales and is not an interval-arithmetic certificate.

For the same counted likelihood, the distinct in-window target $E_3(t_c) - E_0(t_c)$ has local standard-deviation bounds 0.0101 at $t_c = 0.3$ and 0.00176 at $t_c = 0.5$. Its target derivative is $(1, t_c, t_c^2/2)$ in the three diagonal-derivative coordinates. These values quantify the easier later-time target; they are not endpoint error bounds.

Supplemental Note 6. An admissible-prior van Trees calculation

The quadratic S3 class has 24 parameters. Take independent Beta$(3,3)$ densities on $[\theta_{p,*} - w_p, \theta_{p,*} + w_p]$, with $w_p = \sqrt{7}s_p$. For the energy coefficients of orders zero, one and two set $s_p = 0.1$ in their respective dimensionless units. For each corresponding coupling coefficient set $s_p = 0.1g_j(0)$ in its own derivative-coordinate units. These are explicit mathematical prior widths. Each marginal variance is s_p^2 and its prior information is $10/w_p^2$. The densities and their first derivatives vanish at the support boundaries.

Throughout the support all relevant quadratic couplings stay above 0.4197 on $[0,0.24]$. A Gershgorin bound gives $\sup_t \| H(t) \| < 5.763$, hence the integrated norm is less than $1.384 < \pi/2$. The survival-amplitude speed bound gives $|c_0(t)| \geq \cos(0.24 \times 5.763) > 0.1865$, so the projective likelihood chart is uniformly regular.

For independent Gaussian coordinate components with $\sigma = 10^{-8}$ and 61 times in $[0.024,0.24]$, the matrix van Trees inequality uses

$$\mathcal{J}_B = \mathbb{E}_\pi[J(\theta)^T J(\theta)/\sigma^2] + \mathrm{diag}\big(10/w_p^2\big), \mathbb{E}_\pi \mathbb{E}_\theta\big(\hat{E}_3 - E_3\big)^2 \geq e_3^T \mathcal{J}_B^{-1} e_3. \tag{S25}$$

The likelihood-information average is essential. With 200 independent prior draws and seed 20261014, the numerical Bayesian RMSE lower-bound estimate is 1.15×10^{-3}. The two half-sample values are 1.151×10^{-3} and 1.145×10^{-3}. Evaluating only at the central parameter gives 1.23×10^{-3}, about 7% larger. The former is a numerical integration estimate of the prior-averaged bound, while the latter is a pointwise regularized scale. The exact inequality holds for the exact prior average; a finite Monte

Carlo estimate is not a certified lower bound. The empirical 95% bootstrap integration interval is $[1.140,1.156] \times 10^{-3}$ from 2,000 resamples. It measures integration uncertainty, rather than Hamiltonian confidence coverage. The code records the sampled matrices, environment and resampling controls. The prior-regularized matrix has normalized minimum eigenvalue 8.3×10^{-5}, making double-precision inversion adequate for this calculation. The inverse-information plug-in estimate has an upward Jensen bias; its magnitude is not inferred from a single Monte Carlo integration. This Bayesian risk has a different averaging experiment from the flat-nuisance frequentist S3 scale 0.0699.

Supplemental Note 7. Site preparations and oscillator-noise aliases

Fix a finite real Jacobi chain with differentiable coefficients at $t_0 = 0$ and $g_n(0) > 0$. Starting from $c(0) = e_k$, Schrödinger's equation gives $c' = -iHc/\hbar$ and $c'' = -i\dot{H}c/\hbar - H^2c/\hbar^2$. Since $c_j(0) = 0$ for $j \neq k$,

$$w'_{j\leftarrow k} = c'_j, w''_{j\leftarrow k} = c''_j - 2c'_j c'_k. \tag{S26}$$

For adjacent indices, $(H^2)_{jk} = g_{\min(j,k)}(E_j + E_k)$. The formulas in the main text follow immediately. They involve no deeper energies or couplings and require no analyticity.

Selecting every second vertex covers m consecutive edges with $\lceil m/2 \rceil$ site preparations. This count is necessary for identifying arbitrary independent $\dot{g}_n(0)$ from site-prepared second jets, even with complete state readout: $\dot{H}e_k$ contains only coupling derivatives incident to k. If an edge has no prepared endpoint, varying its derivative leaves all available second jets unchanged. Hence the preparation set must be a vertex cover of the path; the matching of $\lceil m/2 \rceil$ disjoint alternate edges proves the lower bound. First jets cannot identify $\dot{H}$ under any preparation. This is a restricted access statement; it makes no claim about arbitrary superpositions.

Assume $H(t)$ preserves photon-number parity and its even block is Jacobi. The odd block may be any Hermitian parity-preserving block. Let $p = 2k$, $q = p + 2$, $h = H_{qp}(0) \neq 0$, and prepare $\rho(0) = |p\rangle\langle p|$. Assume enough regularity and common domains for the first two density derivatives to exist. For a finite oscillator cutoff, require all prepared and probed levels to lie away from its upper boundary so that $a^\dagger a$ and $aa^\dagger$ have their oscillator values on the involved levels. Define $r_p = \kappa_- p + \kappa_+(p + 1)$. Direct differentiation of the dissipative model in the main article gives

$$\rho'_{qp}(0) = -ih/\hbar, \rho'_{pp}(0) = -r_p. \tag{S27}$$

A loss or excitation jump changes parity. Thus its recycling term contributes no qp coherence at second order: two jumps acting on the initial diagonal state are diagonal, while one jump acting on its first Hamiltonian derivative produces an odd-block coherence. Number dephasing acts on the qp coherence with eigenvalue $-\gamma(q-p)^2/2 = -2\gamma$. The Hamiltonian part of the second derivative has contributions from $\dot{h}$ and $(E_q - E_p)h$. The dissipative contributions are $-[r_p + (r_p + r_q)/2 + 2\gamma]\rho'_{qp}(0)$, yielding

$$\frac{\rho''_{qp}(0)}{\rho'_{qp}(0)} = \frac{\dot{h}}{h} - \frac{i}{\hbar}(E_q - E_p) - \frac{3r_p + r_q}{2} - 2\gamma. \tag{S28}$$

Since $\rho_{pp}(0) = 1$ and $\rho_{qp}(0) = 0$, the ratio rule gives

$$\frac{w_k''(0)}{w_k'(0)} = \frac{\rho''_{qp}(0)}{\rho'_{qp}(0)} - 2\rho'_{pp}(0) = \frac{\dot{h}}{h} - \frac{i}{\hbar}(E_q - E_p) + \frac{r_p - r_q}{2} - 2\gamma. \tag{S29}$$

Now $(r_p - r_q)/2 = -(\kappa_- + \kappa_+)$, proving the initial rate identity in the main article. Time derivatives of the dissipative rates act on the initial diagonal state and contribute no qp coherence, so the formula also applies to smooth time-dependent rates evaluated at zero. The same real-part identity holds for complex h: its Hamiltonian contribution is $\mathrm{Re}(\dot{h}/h) = d\log|h|/dt$. The imaginary part then also contains the coupling phase velocity.

For an additional noise term $\gamma_f \mathcal{D}[f(N)]$ with real f, the rate shift is $-\gamma_f[f(p+2) - f(p)]^2/2$. It cancels between all edges when $f(N)$ is affine in N, and generally survives for nonlinear f. This provides an explicit limitation of the differential diagnostic.

Vacuum aliases at arbitrary depth

For the pair-driven oscillator with number-diagonal energy and banded even and odd pair ladders, a leading vacuum-to-$2n$ coherence requires n Hamiltonian steps. Loss or gain recycling that returns to this even coherence requires two jumps and a higher-occupation coherence, or additional propagation after a diagonal return, and therefore occurs beyond the first two onset orders. At these orders the normalized coherence damping is

$$-n(\kappa_- + \kappa_+) - 2n^2\gamma. \tag{S30}$$

The reference normalization removes its common population decay. In the next-order recursion this damping shifts the increment $R_n - R_{n-1}$ by the displayed expression; R_n accumulates those increments, so $\mathrm{Re}(R_n - R_{n-1})/n$ shifts by $-\kappa_- - \kappa_+ - 2n\gamma$. The same reasoning applies to smooth time-dependent rates evaluated at preparation. The vacuum claim uses the banded pair ladders, whereas the adjacent-site second-jet proof allows an arbitrary parity-preserving odd block. A sixteen-Fock-level density-series calculation verifies four vacuum layers with residual below 3×10^{-16}. Independent initial-density derivatives for 200 random parity-preserving generators verify the adjacent-site identity with residual below 1.9×10^{-14}. Cutoff boundaries must remain above every level involved in the derivative calculation.

Two-photon loss and diagonal preparation mixtures

For the additional term $\kappa_2\mathcal{D}[a^2]$, define $d_p = p(p-1)$. With preparation $|p\rangle$, $q = p + 2$, the anti-commutator contributions and reference normalization give an additional shift $\kappa_2(d_p - d_q)/2 = -\kappa_2(2p+1)$ in $\mathrm{Re}(w''/w')$. The recycling contribution to the qp coherence would require the initially absent higher pair coherence; cross terms that recycle an initial population down by two cannot return to qp in one Hamiltonian step. Hence at $p = 2k$ the shift is $-\kappa_2(4k+1)$, retaining $-8\kappa_2$ between $k = 2$ and $k = 0$. For vacuum and the banded pair ladder, the damping entering the nth onset recursion is $-\kappa_2(2n)(2n-1)/2$. Its contribution to the increment $R_n - R_{n-1}$ divided by n is $-\kappa_2(2n-1)$. An eighteen-Fock-level density recurrence for four vacuum layers and forty random parity-preserving generators verifies the vacuum and site shifts within 1.8×10^{-15}. The arbitrary-odd-block site check does not extend the loss/gain vacuum claim beyond its banded hypothesis.

Under unitary real-Jacobi evolution, an initial diagonal mixture with populations $p_k \neq p_j$ on an adjacent pair has

$$w' = -\frac{ig(p_k - p_j)}{\hbar p_k}, \frac{w''}{w'} = \frac{\dot{g}}{g} - \frac{i(E_j - E_k)}{\hbar}. \tag{S31}$$

The initial diagonal population derivative vanishes. Thus the exact second-jet logarithmic rate is unchanged by this incoherent mixture, whereas an inferred coupling is rescaled by $(p_k - p_j)/p_k$. A two-percent neighbouring population gives factor $0.96/0.98 \simeq 0.9796$ on the seeded edge. Finite-

window fits can still acquire different truncation biases, as shown below. The statement concerns diagonal preparation under unitary evolution.

Supplemental Note 8. Acquisition and preparation mismatch

Deterministic onset and offset controls

These controls use 61 times in $[0.024,0.24]$, a degree-$n+2$ local polynomial fit with ideal endpoint onset powers, and no observation noise. They therefore isolate model/preparation mismatch from sampling fluctuations.

Table S4. Noiseless fitted third-layer quantities under specified acquisition mismatch. The instantaneous target is $(g_2, E_3 - E_0, \dot{g}_2/g_2) = (0.58,0.70,0.2414)$.

Model	g_2	$E_3 - E_0$	$\dot{g}_2/g_2$
Ideal onset	0.5798	0.7007	0.2504
Real ratio offset 10^{-6}	0.5807	−0.6705	0.2604
Real ratio offset 10^{-5}	0.6254	−11.62	−1.031
Real ratio offset 10^{-3}	168.1	−17.11	−69.90
Coupling-only ramp, $\tau_r = 0.002$	0.5642	0.8533	0.6211
Geometric thermal Fock state, $\bar{n} = 0.01$	0.5797	0.7007	0.2504
One-percent incoherent site-1 population	0.5738	0.7014	0.2530

A residual time-dependent readout phase supplies a separate energy alias. If $\tilde{z}_n = e^{i\alpha_n(t)} z_n$, the endpoint ratio changes by $\tilde{R}_n - R_n = i(n+1)\dot{\alpha}_n(0)$. Thus the inferred energy offset shifts by $-\hbar[(n+1)\dot{\alpha}_n - n\dot{\alpha}_{n-1}]$, with $\alpha_0 = 0$. More copies do not remove this systematic phase-rate error.

The offset adds the same real constant to z_1, z_2, z_3 before fitting and is not subtracted. The ramp replaces each off-diagonal coupling by $g_j(t)\left(1 - e^{-t/\tau_r}\right)$ while retaining the diagonal law. It violates the positive-onset hypothesis, even though the fractional drive change inside the window is at most $e^{-12} \simeq 6.14 \times 10^{-6}$. The thermal control uses a twelve-Fock-level density matrix with initial probabilities $p_m = (1+\bar{n})^{-1}[\bar{n}/(1+\bar{n})]^m$, normalized at the truncation, the prescribed even-sector Hamiltonian and zero odd-sector block. The measured ratios are $\rho_{2n,0}/\rho_{00}$. The final row instead uses

$\rho(0) = 0.99|0\rangle\langle 0| + 0.01|2\rangle\langle 2|$ in the even sector. This is an incoherent population error, distinct from the geometric thermal distribution. The small geometric-thermal effect is specific to that state; arbitrary mixed preparation is not covered. The ideal polynomial fit itself has rate bias, so the small thermal change should be compared with the ideal row.

Two-site finite-window mismatch controls

For the same six-level chain, pure preparations at sites 0 and 2 and cubic fits of the three incident ratios give the following noiseless controls. The true endpoint energy is 0.70 and the true rate difference $\chi_2 - \chi_0$ is 0.50165. Energies are assembled as $(E_1 - E_0) - (E_1 - E_2) + (E_3 - E_2)$ from the incident pairs.

Table S5. Two-site cubic-fit systematics, 61 times in $[0.024,0.24]$. Coherent rows show phase-sweep ranges. These fitted errors are separate from the local Fisher bounds.

Model	Estimated $E_3 - E_0$	Estimated $\chi_2 - \chi_0$
Ideal, fixed zero intercept	0.7160	0.4928
Ideal, free intercept	0.7249	0.4926
Common ramp, $\tau_r = 0.002$	0.7585	0.5179
Site-2 ramp 10% slower	0.7613	0.6196
Real ratio offset 10^{-6}	0.7170	0.4928
Real ratio offset 10^{-5}	0.7258	0.4933
Common envelope, matched ramp	0.7601	-0.00725
Common envelope, mismatched ramp	0.7629	0.0942
Coherent $\varepsilon = 0.01$, zero intercept	$[-16.4,17.8]$	$[-1.01,5.99]$
Coherent $\varepsilon = 0.01$, free intercept	$[0.714,0.736]$	$[0.470,0.516]$
Coherent $\varepsilon = 0.03$, zero intercept	$[-67.7,65.6]$	$[-9.07,62.0]$
Coherent $\varepsilon = 0.03$, free intercept	$[0.690,0.756]$	$[0.424,0.561]$
Coherent $\varepsilon = 0.10$, zero intercept	$[-6.67,8.66]$	$[-0.438,1.32]$
Coherent $\varepsilon = 0.10$, free intercept	$[0.587,0.822]$	$[0.264,0.721]$
Incoherent 2%, zero intercept	0.7154	0.4943
Incoherent 2%, free intercept	0.7240	0.4949

Both ideal fits are biased relative to the true values 0.70 and 0.50165. The common-envelope rows use $g_j(t) = g_j(0)(1 + 0.2t + 0.02t^2)$ with the same diagonal law, so their true rate difference is zero; their unramped fitted difference is -0.00189. The shared-ramp residual therefore depends on the model. Additive offsets 10^{-6}–10^{-5} have much smaller effects here than in the endpoint fit of Table S4.

For coherent controls, each preparation is $\sqrt{1-\varepsilon^2}|k\rangle + \varepsilon e^{i\varphi}|k+1\rangle$, $k = 0,2$, with a common phase on a 64-point uniform grid. A free intercept fits the seeded zeroth-order coherence and substantially reduces error. It does not remove the preparation-induced changes in the true derivatives. For example,

exact second-jet inference at $\varepsilon = 0.03$ itself gives energy range $[0.666, 0.731]$ and rate-difference range $[0.436, 0.566]$. The apparently improved fit is therefore conditional on this specified admixture; calibrated preparation remains necessary. Incoherent rows use the corresponding diagonal two-percent neighbouring mixture. Its exact unitary second-jet rate is invariant, while finite-window truncation bias changes slightly.

Exact initial-jet cancellation does not imply cancellation of every finite-window fitting error. A coupling-only zero-onset ramp violates the positive-onset assumptions. Matching preparation-dependent drive onset and calibrating pair-readout offsets remain requirements for physical use.